

- 7 A circuit contains a power supply that provides a sinusoidal alternating input voltage V_{IN} . There is an output voltage V_{OUT} across a load resistor R , as shown in Fig. 7.1.

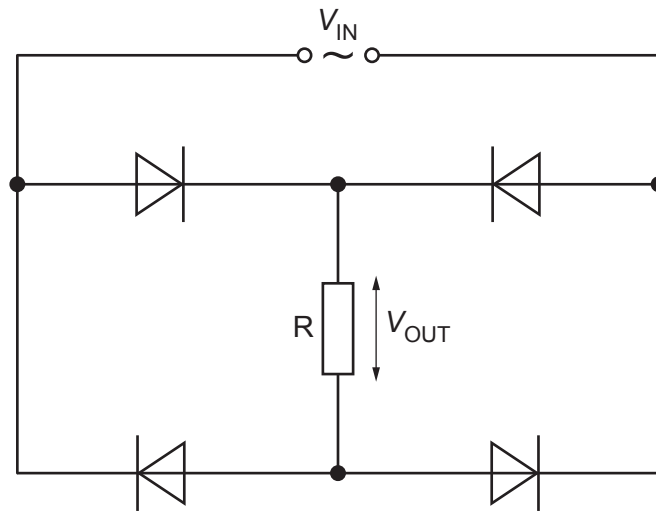


Fig. 7.1

- (a) State the purpose of the circuit in Fig. 7.1.

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..... [2]

- (b) Fig. 7.2 shows the variation of V_{OUT} with time t .

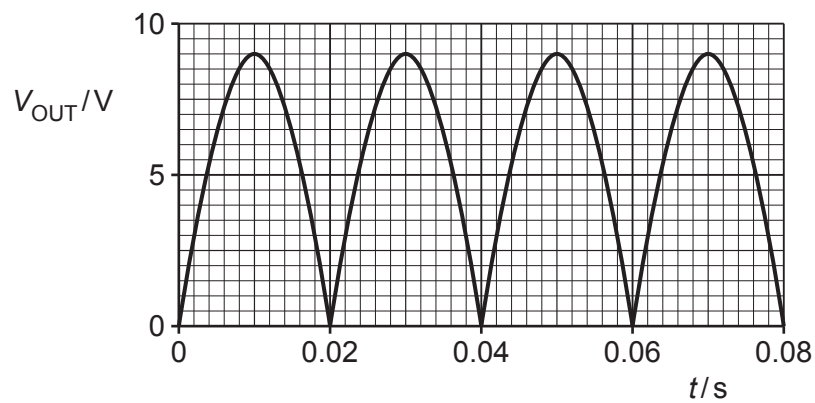


Fig. 7.2

- (i) The load resistor R has a resistance of $370\ \Omega$.

Show that the maximum power dissipated in R is 0.22 W .

[2]

- (ii) On Fig. 7.3, sketch the variation with t of the power P dissipated in R .

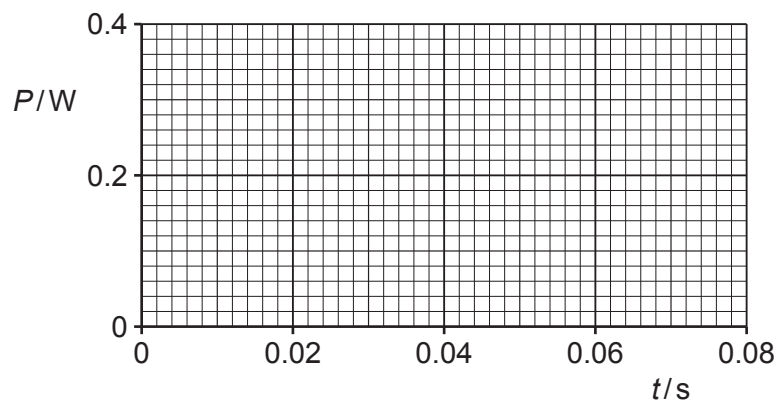


Fig. 7.3

[3]

- (iii) Calculate the mean power dissipated in R .

mean power = W [1]

- (c) The circuit of Fig. 7.1 is disconnected, and R is connected directly across the power supply.

Explain, without calculation, how the mean power now dissipated in R compares with the answer in (b)(iii).

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 [2]

[Total: 10]